
Lift, See, Act: Hierarchical Robot Policy Pretraining with 3D Foundation Models

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Abstract

Robot policy pretraining based on human videos is crucial for improving the policy’s generalization ability. One main challenge for this task is the lack of explicit action-relevant representations in such unlabeled data. Recent works tend to estimate 3D hand motion trajectories from videos using 3D foundation models (3D FMs). However, they keep solely the sparse trajectories as supervision, ignoring the rich, fine-grained information produced during the trajectory extraction process. To address this issue, we propose **LSA**, a hierarchical robot policy pretraining framework that includes three phases: **Lift**, **See**, and **Act**. Specifically, the **Lift** phase elevates 2D video observations into dense 3D representations by adaptively aligning features in the policy’s shallow layers with the intermediate representations extracted from multiple 3D FMs. Building upon these lifted representations, the **See** phase equips the policy with explicit geometric and interactive awareness through dual guidance. It introduces depth-map reconstruction to help the model comprehend 3D spatial layouts and utilizes hand region cues to explicitly supervise the encoder’s attention toward task-relevant interaction hotspots. Empowered by the dense 3D representation learning and precise spatial guidance, our policy achieves robust and accurate robotic manipulation in the **Act** phase. Extensive experiments on diverse simulation and real-world tasks demonstrate that LSA significantly outperforms current state-of-the-art approaches. The code of this work will be released soon.

1 Introduction

Large-scale pretraining is crucial for improving the generalization ability of machine learning models [1, 2]. In robotics, pretraining is equally important for policy learning, since high-quality manipulation data are expensive to collect and therefore difficult to scale up. Human videos are a key source of pretraining data, as they cover a wide range of real-world manipulation scenarios and contain rich manipulation knowledge [3, 4]. However, using human videos to pretrain robot policies remains a challenge, as such data inherently lack action-relevant representations [5].

Existing methods for extracting explicit action-relevant representations from human videos can be divided into two categories. The first category of methods extracts 2D visual primitives from video frames, such as contact points [6, 7], contact heatmaps [8], and object bounding boxes [9], to characterize where contact occurs and which regions are potentially involved in manipulation. However, these methods lack temporal representations of the 3D manipulation process. The second category of methods attempts to estimate 3D hand motion trajectories from videos [10–14]. These trajectories reflect both the displacement of the hand in the 3D space and the temporal evolution of that displacement; therefore, they are more in line with the robot’s action prediction space. Recently, with the development of 3D foundation models (3D FMs) [15–18], the accuracy of the estimation of 3D hand motion trajectories has improved substantially [12–14].

However, as illustrated in Fig. 1, the latter category of approaches only keeps the estimated hand motion trajectories as supervision for robot policy pretraining. They ignore more fine-grained knowledge, e.g., rich geometric representations and depth maps, produced by the 3D FMs during the hand motion trajectory prediction process. This dense knowledge is fundamental for a robot policy in predicting the 3D motion of the end effector. This insight motivates us to study how to make the best use of multi-tiered knowledge from 3D FMs to promote robot policy pretraining.

Consequently, we propose **LSA**, a hierarchical framework for robot policy pretraining based on three phases: **Lift**, **See**, and **Act**. Specifically, in the **Lift** phase, our goal is to lift 2D video observations into rich 3D representations. We first construct a 3D hand motion trajectory extraction pipeline utilizing multiple 3D FMs to process unlabeled human videos. This pipeline not only extracts the motion trajectories, but also keeps dense intermediate representations and depth maps produced by these 3D FMs. We then enforce the shallow-layer representations of the policy to align with the intermediate representations of each 3D FM, respectively. We also propose a strategy to adaptively adjust the weights of these alignment losses. This process effectively lifts the policy’s 2D representations into 3D, endowing the policy with more fine-grained geometric knowledge.

Building upon these lifted representations, the **See** phase equips the policy with explicit geometric and interactive awareness. While the Lift phase establishes a dense 3D representation space, these representations are implicit. Robot manipulation inherently requires an intuitive and precise understanding of the structure of the physical scene, along with a distinct visual focus on the manipulation objectives. To achieve this, the See phase provides dual guidance: explicit structural guidance and attentive guidance. First, leveraging the extracted depth maps, we introduce depth map reconstruction as an auxiliary task, providing explicit 3D geometric constraints to enable the model to comprehend 3D spatial layouts. Second, we utilize the hand region as a cue to guide the encoder’s attention toward task-relevant interaction hotspots. Specifically, we encode 2D hand keypoints into a Gaussian heatmap to represent the hand region. We then select the attention map most similar to this heatmap across all heads and layers as our target, and finally use it to supervise the mean attention of all heads within a layer. The above dual guidance ensures the policy explicitly **Sees** the 3D scene structure and concentrates on the task-relevant interaction regions.

To the best of our knowledge, LSA is the first approach to leverage multi-tiered 3D information extracted from 3D foundation models for hierarchical robot policy pretraining on human videos. Empowered by the dense 3D representation learning in the Lift phase and the dual guidance in the See phase, our policy predicts 3D hand motion trajectories in the **Act** phase, ultimately achieving robust and accurate robot manipulation. Extensive experiments demonstrate the effectiveness of our design and show that LSA outperforms state-of-the-art methods on various simulation and real-world tasks.

2 Related Works

Existing approaches to pretraining robot policies from human videos often construct action-relevant representations to bridge the gap between passive visual observations and robot control. Based on the space in which these representations are extracted, existing approaches can be broadly categorized into 2D and 3D methods.

2D Methods. The first category of methods constructs action-relevant representations within the 2D image space, which can be further divided into implicit and explicit methods. Implicit methods model

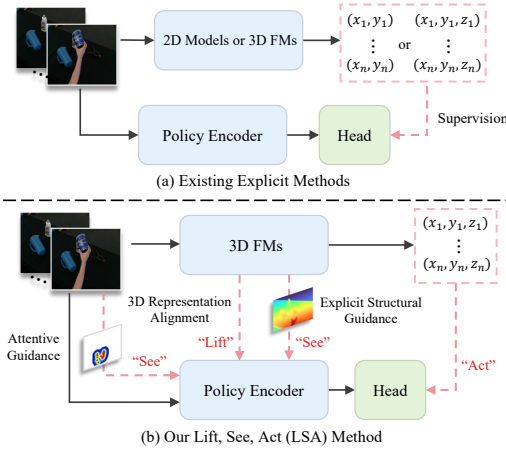


Figure 1: Comparisons between methods that extract explicit action-relevant representations from human videos. (a) Existing methods adopt solely the estimated hand motion trajectories as supervision [8, 12, 10]. (b) Our LSA leverages multi-tiered information from multiple 3D FMs to facilitate robot policy pretraining.

inter-frame relationships to extract abstract state representations. One line of work uses contrastive learning [19, 20] to constrain the distance between visual embeddings across different time periods, capturing the temporal dynamics of visual observations (i.e., how the current state transitions to the next). To characterize these state transitions more precisely, other approaches learn discrete latent actions between frames through VQ-VAE objectives [21], subsequently using these latent actions as pseudo-action labels for unannotated videos [3, 22, 23]. Recently, some methods have trained a video prediction model [24–26, 4] to extract predictive visual representations that encode current and future frames, providing future valuable information for embodied agents.

In contrast, explicit methods directly extract 2D visual primitives at the pixel level. These approaches typically identify contact points [6, 7], contact heatmaps [8], or bounding boxes [9, 27] to delineate potential interaction regions. Furthermore, extracting human hand poses from video streams serves as a simple but scalable approach to represent the interaction trajectories [7, 28].

3D Methods. The above 2D representations fundamentally lack explicit depth information, limiting their ability to capture inherently 3D physical interactions. To address this limitation, the second category of methods turns to explicit 3D action-relevant representations, such as 3D hand trajectories [12] and parametric hand poses (e.g., MANO [29]) [30, 6]. Due to their 3D nature, these representations align more tightly with the robotic action space.

These 3D methods can be further divided into two groups according to their data requirements. The first group uses videos equipped with annotated 3D hand poses and camera parameters, typically captured through AR/VR devices or RGB-D SLAM [31]. They directly adopt these ground-truth 3D hand poses as supervision for the pretraining of robot policies [30, 10, 32, 33]. However, these videos are recorded in relatively controlled settings, which affects the scale of the video databases. In comparison, the second group constructs 3D representations directly from unlabeled, in-the-wild videos. Early efforts employ structure-from-motion and depth estimation models to recover camera intrinsics and metric depth. They then combine these estimations with continuous 2D hand poses to obtain 3D hand motion trajectories [11, 34]. Recently, the rapid advancement of 3D Foundation Models (FMs) [15, 17, 16] has enabled the extraction of 3D hand trajectories with substantially improved fidelity and efficiency [12, 10].

In this paper, we also rely on 3D FMs to construct action-relevant representations. Unlike existing methods that focus exclusively on 3D hand trajectories, we extract hierarchical priors from the 3D FM’s inference process, enabling more robust robot policy pretraining.

3 Methods

3.1 Problem Formulation and LSA Overview

Following prior work [8, 19, 6, 35, 36], we pretrain the visual encoder part of the robot policy using large-scale human videos. Once trained, the encoder, denoted $\mathcal{F}_{\phi^*}$, is frozen and serves as a perceptual front-end for robot policy. Given a current visual observation $\mathbf{I}$, the encoder maps it to a latent embedding $\mathbf{z} = \mathcal{F}_{\phi^*}(\mathbf{I})$, where ϕ^* represents the fixed parameters of the pretrained encoder. This embedding $\mathbf{z}$ is then concatenated with the proprioceptive state of the robot $\mathbf{s}$ and fed into a policy head π_{θ} to generate an action $\hat{\mathbf{a}} \sim \pi_{\theta}(\cdot \mid \mathbf{z}, \mathbf{s})$. The policy is trained through behavior cloning [37] using a dataset of expert demonstration trajectories $\mathcal{D}_{\text{IL}} = \{\tau_i\}_{i=1}^K$, where each trajectory is defined as $\tau_i = \{(\mathbf{I}_t, \mathbf{s}_t, \mathbf{a}_t)\}_{t=1}^T$. The training objective minimizes the loss of behavior cloning between the predicted action $\hat{\mathbf{a}}_t$ and the ground-truth expert action $\mathbf{a}_t$:

$$\min_{\theta} \mathbb{E}_{(\mathbf{I}_t, \mathbf{s}_t, \mathbf{a}_t) \sim \mathcal{D}_{\text{IL}}} [\mathcal{L}_{\text{BC}}(\hat{\mathbf{a}}_t, \mathbf{a}_t)].$$

During evaluation, the policy is deployed in a closed-loop control setting and its performance is quantified by the success rate averaged over multiple deployments.

As highlighted in Section 1, relying solely on estimated hand trajectories for robot policy pretraining wastes the fine-grained geometric knowledge inherently produced by 3D Foundation Models (3D FMs). To make the best use of this multi-tier knowledge, we propose **LSA**, a hierarchical pretraining framework consisting of three distinct phases: **Lift**, **See**, and **Act**. First, the **Lift** phase elevates 2D video observations into a dense 3D representation space by aligning the policy’s shallow-layer features with intermediate representations from 3D FMs. Building upon this, the **See** phase equips the policy with explicit geometric and interactive awareness through structural and attentive dual

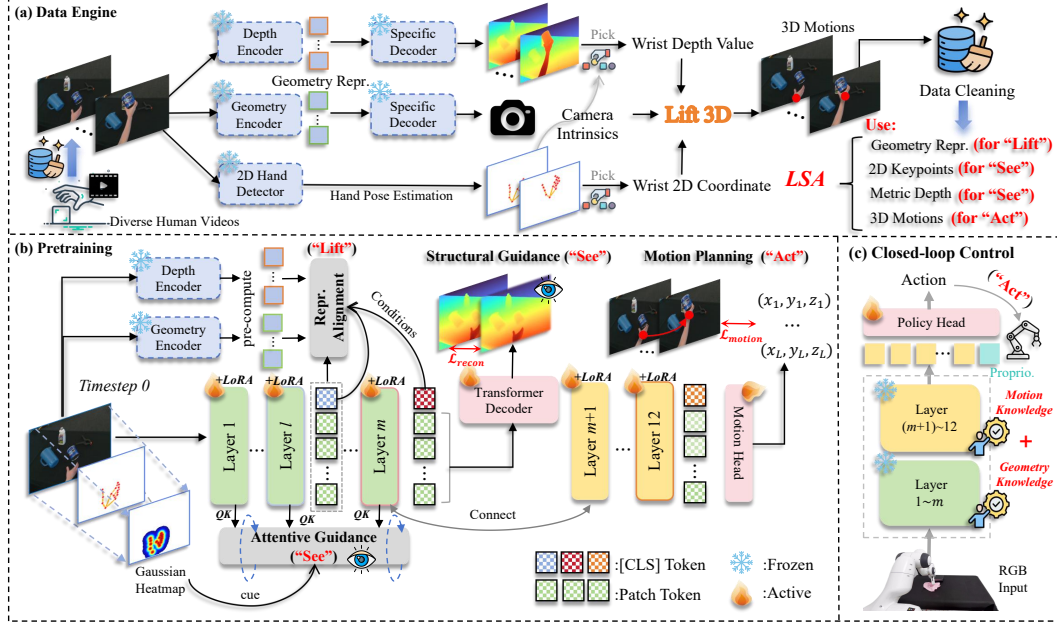


Figure 2: Architecture overview of LSA. (a) In the data engine phase, we utilize two 3D FMs to extract depth maps and camera intrinsics, respectively. Combined with the extracted 2D wrist coordinates, we obtain 3D hand motion trajectories. Furthermore, we also extract dense geometric representations and depth maps. (b) In the pretraining phase, we first align the policy’s shallow-layer representations with the intermediate representations of the two 3D FMs pre-computed during the data engine phase. Then, we introduce depth-map reconstruction to equip the model with precise structural knowledge, and utilize hand heatmaps to guide the model’s attention toward interaction hotspots. Following this, we predict the 3D hand motion trajectories. (c) Finally, we freeze the pretrained LSA and combine it with a policy head to perform closed-loop robotic control.

guidance. Finally, empowered by these representations, the **Act** phase translates visual inputs into 3D hand motion trajectories for robust manipulation.

3.2 The Lift Phase: 3D Representation Elevation

Data Engine. To extract 3D hand motion trajectories, we develop a data engine using multiple 3D FMs. Given an input video frame, we first extract the 2D wrist coordinates $\mathbf{p}_w$. Subsequently, we employ VGGT [15] to estimate camera intrinsics $\mathbf{K}$, and Depth Pro [16] to predict a metric depth map $\mathbf{D}_g$. During this process, the two 3D FMs generate rich intermediate geometric representations, denoted as $\mathbf{E}_c$ and $\mathbf{E}_d$, respectively. Using $\mathbf{K}$ and $\mathbf{D}_g$, we lift the 2D wrist coordinates $\mathbf{p}_w$ into the 3D space. Finally, we conduct necessary data cleaning to obtain high-quality pretraining data. Details of data cleaning and 3D lifting are provided in Appendix A.4.

Dense Representation Alignment. Given an input image $\mathbf{I}_h \in \mathbb{R}^{H \times W \times 3}$, the vision encoder extracts visual features from it. Let $\mathbf{X}_a \in \mathbb{R}^{N_a \times D_a}$ denote the visual tokens from the l -th layer of the encoder. To distill the fine-grained geometric knowledge of the two 3D FMs into the policy, we align the shallow-layer visual tokens $\mathbf{X}_a$ with their geometric representations $\mathbf{E}_c \in \mathbb{R}^{N_c \times D_c}$ and $\mathbf{E}_d \in \mathbb{R}^{N_d \times D_d}$, respectively.

To ensure spatial resolution compatibility, we first reduce the number of tokens in $\mathbf{E}_c$ and $\mathbf{E}_d$ to match N_a via downsampling and add positional encodings to their respective image patch tokens. We denote the resulting spatially-aligned features as $\mathbf{E}'_c \in \mathbb{R}^{N_a \times D_c}$ and $\mathbf{E}'_d \in \mathbb{R}^{N_a \times D_d}$. Next, to ensure compatibility with the feature dimensions, we map the embedding dimension of $\mathbf{X}_a$ to D_c and D_d using two separate MLPs, respectively. Following the typical token-level alignment paradigm [38], we employ a cosine similarity score to maximize the alignment between the visual tokens and the geometric signals. To balance the contributions of multiple geometric features, we introduce a set of

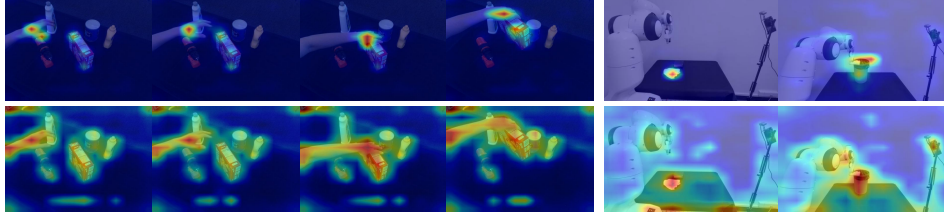


Figure 3: Effects of attentive guidance (AG). Left: attention maps on human videos. Right: attention maps on robotic scenes. In both cases, the top row shows results w/ AG and the bottom row shows results w/o AG.

dynamic modulation weights w_c and w_d . The overall alignment loss is formulated as:

$$\mathcal{L}_{\text{align}} = -\frac{1}{N_a} \sum_{i=1}^{N_a} [w_c \cdot \mathcal{S}(\text{MLP}_c(\mathbf{x}_{a,i}), \mathbf{e}'_{c,i}) + w_d \cdot \mathcal{S}(\text{MLP}_d(\mathbf{x}_{a,i}), \mathbf{e}'_{d,i})], \quad (1)$$

where $\mathbf{x}_{a,i}$ denotes the i -th visual token in $\mathbf{X}_a$, and $\mathbf{e}'_{c,i}$ and $\mathbf{e}'_{d,i}$ represent the tokens in $\mathbf{E}'_c$ and $\mathbf{E}'_d$ that spatially correspond to $\mathbf{x}_{a,i}$, respectively. $\mathcal{S}(\cdot, \cdot)$ denotes the cosine similarity function. This alignment process effectively lifts the policy’s 2D representations into 3D, endowing it with fine-grained geometric awareness.

To ensure that the model captures valuable geometric knowledge from multiple teachers, we propose a dynamic weight assignment mechanism inspired by [39]. We derive a condition query by aggregating the [CLS] tokens $\mathbf{c}_l$ and $\mathbf{c}_m$ from the l -th layer and the m -th layer of the vision encoder, where the m -th layer is further used for depth reconstruction. Then, we compute the scaled dot-product similarity between this query and the geometric tokens to adaptively determine the dynamic weights w_c and w_d for $\mathbf{E}'_c$ and $\mathbf{E}'_d$, respectively. More details are provided in **Appendix A.2**.

3.3 The See Phase: Dual Attentive and Structural Guidance

Although the Lift phase establishes a dense 3D representation space, these features remain largely implicit. Robot manipulation inherently requires an intuitive understanding of the structure of the physical scene, alongside a distinct visual focus on the manipulating agent and the target object. To achieve this, the See phase provides dual guidance.

Attentive Guidance. Learning affordance knowledge from human videos benefits policy pretraining [8, 6]. To further improve the policy’s perception of hand-centric interaction regions, we design an attentive guidance mechanism. Specifically, we first warm up the model for several epochs with the motion planning objective, allowing initial task-relevant attention patterns to emerge (see **Appendix A.3**). Given the detected hand keypoints, we generate a Gaussian heatmap and downsample it to the patch resolution, obtaining a spatial hand prior $\mathbf{h} \in \mathbb{R}^{N_a}$, where N_a denotes the number of spatial patches. For each head in the first R layers, we extract the [CLS]-to-patch attention map $\mathbf{a}_{n,k} \in \mathbb{R}^{N_a}$, where n and k denote the layer and head indices. We then compute its alignment score with the hand prior as:

$$s_{n,k} = \frac{1}{N_a} \sum_{i=1}^{N_a} \mathbf{a}_{n,k,i} \mathbf{h}_i. \quad (2)$$

The attention map with the highest score is selected as the guidance target:

$$(n^*, k^*) = \arg \max_{n,k} s_{n,k}, \quad \mathbf{a}^* = \mathbf{a}_{n^*,k^*}. \quad (3)$$

We then use $\mathbf{a}^*$ as a fixed self-distillation target to refine the averaged shallow attention. For each layer $n \leq R$, we compute the averaged multi-head attention map $\bar{\mathbf{a}}_n$ and minimize its distance to $\mathbf{a}^*$ after spatial normalization:

$$\mathcal{L}_{\text{att}} = \sum_{n=1}^R \|\phi(\text{sg}(\mathbf{a}^*)) - \phi(\bar{\mathbf{a}}_n)\|_2^2, \quad (4)$$

where $\phi(\cdot)$ denotes Min-Max normalization and $\text{sg}(\cdot)$ denotes stop-gradient. In this way, attentive guidance transfers the model’s intrinsic hand-centric attention pattern to the shallow layers, enabling them to better focus on task-relevant hand-object regions.

As shown in Fig. 3, we visualize the averaged attention maps over the first R shallow layers to analyze the effect of attentive guidance. After applying attentive guidance, the model shows a clearer focus on the hand region. Notably, it also exhibits emergent attention to interacting objects, which can be attributed to the motion planning (Sec. 3.4) objective that tightly couples future hand trajectories with hand-object spatial relations. Moreover, Fig. 3 shows that these refined attention patterns transfer to robotic manipulation scenes, enabling the model to focus more on task-relevant regions while reducing attention to background.

Explicit Structural Guidance. To provide explicit 3D geometric constraints and enable the model to comprehend 3D spatial layouts more precisely, we introduce depth-map reconstruction as an auxiliary task. We utilize the image patch tokens from the m -th layer of the encoder, denoted as $\mathbf{X}_m$, which are processed by a transformer decoder to regress the depth map. This reconstruction is supervised by the metric depth map $\mathbf{D}_g$ generated during the data engine stage, and the reconstruction loss $\mathcal{L}_{\text{recon}}$ is computed using the standard ℓ_2 distance.

3.4 The Act Phase: Motion Planning

Empowered by the dense 3D representation learning in the Lift phase and the precise dual guidance in the See phase, the Act phase predicts the 3D hand motion trajectory. For a training sample with $L + 1$ sequential frames, we first lift the wrist locations into 3D space using the estimated camera intrinsics and the metric depth map, and construct the ground-truth future trajectory $\mathbf{T} \in \mathbb{R}^{L \times 3}$ from the subsequent L frames. We then average-pool the final-layer visual tokens of the initial frame into a global visual embedding, which is fed into a single-layer MLP prediction head to predict the future trajectory $\hat{\mathbf{T}} \in \mathbb{R}^{L \times 3}$. Finally, the motion planning objective $\mathcal{L}_{\text{motion}}$ is optimized by minimizing the ℓ_1 distance between $\hat{\mathbf{T}}$ and $\mathbf{T}$. By grounding action prediction in these rich 3D priors and refined attention, the policy acquires robust, generalized motion planning capabilities, establishing a strong foundation for downstream robot control.

4 Simulation Experiments

4.1 Setup

Evaluation Setup. We evaluate LSA on three simulation benchmarks: MetaWorld [40], DexArt [41], and Adroit [42]. MetaWorld evaluates tabletop manipulation with a Sawyer arm and gripper, while DexArt and Adroit focus on dexterous manipulation with multi-fingered hands. For MetaWorld, we use eight tasks covering different difficulty levels: *hammer*, *dial turn*, *assembly*, *hand insert*, *pick place wall*, *shelf place*, *disassemble*, and *stick pull*. For DexArt, we evaluate on *bucket* and *toilet*; for Adroit, we evaluate on *door*, *hammer*, and *pen*. Following prior methods, the pretrained visual encoder is frozen and combined with a lightweight behavior cloning policy head. We report success rates averaged over three random seeds. Detailed data collection, training, and evaluation protocols are provided in **Appendix B.1**.

Baselines. We compare LSA with three groups of baselines. First, we include general-purpose visual representations, including CLIP [43] and DINOv3 [44]. Second, we compare with representative robot visual representation pretraining methods, including MCR [45], VIP [20], R3M [19], VC-1 [46], HRP [8], MPI [9], and R3M-A [36]. Third, we include baselines that directly take 3D point clouds as inputs, including DP3 [47] and KAN-We-Flow [48]. For VIP, we additionally provide cropped target images following its original protocol. More baseline details are provided in **Appendix B.2**.

4.2 Results

As shown in Table 1, LSA obtains consistently strong results across MetaWorld, DexArt, and Adroit. On MetaWorld, LSA performs particularly well on challenging tasks such as *pick-place-wall* and *disassemble*, where accurate spatial perception is critical. On DexArt and Adroit, LSA also achieves competitive or best performance on articulated-object and dexterous manipulation tasks. Compared with direct 3D policy baselines such as DP3 [47] and KAN-We-Flow [48], LSA attains these results without using explicit 3D observations during downstream policy learning, suggesting that the proposed pretraining effectively injects useful 3D-aware representations into the visual encoder. We further evaluate robustness by reducing image brightness by 40% and adding random

Table 1: Success rate (%) reported as mean $\pm$ standard deviation over 3 random seeds on MetaWorld, DexArt, and Adroit. * denotes that cropped target images are used as additional inputs. Task abbreviations: HA (Hammer), DT (Dial Turn), AS (Assembly), HI (Hand Insert), PPW (Pick Place Wall), ShP (Shelf Place), DI (Disassemble), StP (Stick Pull), BU (Bucket), TO (Toilet), DR (Door), HA-A (Adroit Hammer), and PE (Pen). Bold and underline denote the best and second-best results, respectively.

Method	MetaWorld								DexArt		Adroit		
	HA	DT	AS	HI	PPW	ShP	DI	StP	BU	TO	DR	HA-A	PE
<i>Direct 3D Policy</i>													
DP3 (RSS'24)	91.1 $\pm$ 2.8	84.9 $\pm$ 2.0	51.1 $\pm$ 3.3	33.8 $\pm$ 4.0	32.9 $\pm$ 3.0	48.4 $\pm$ 4.7	40.4 $\pm$ 3.5	39.1 $\pm$ 4.0	26.7 $\pm$ 1.7	71.7 $\pm$ 2.2	63.3 $\pm$ 1.7	100.0 $\pm$ 0.0	40.6 $\pm$ 1.0
KAN-We-Flow (ICRA'26)	89.3 $\pm$ 1.4	89.8 $\pm$ 2.8	61.8 $\pm$ 4.1	58.7 $\pm$ 1.4	68.0 $\pm$ 4.0	70.2 $\pm$ 2.8	68.9 $\pm$ 2.0	66.7 $\pm$ 4.0	31.1 $\pm$ 2.6	<u>76.7</u> $\pm$ 1.7	<u>80.5</u> $\pm$ 2.5	100.0 $\pm$ 0.0	<u>56.7</u> $\pm$ 3.4
<i>2D Pretrain Policy</i>													
CLIP (ICML'21)	95.7 $\pm$ 1.2	78.3 $\pm$ 4.6	88.7 $\pm$ 4.0	61.7 $\pm$ 5.7	<u>92.3</u> $\pm$ 1.2	72.3 $\pm$ 5.7	82.7 $\pm$ 3.2	70.3 $\pm$ 3.1	58.7 $\pm$ 3.5	42.0 $\pm$ 0.0	36.7 $\pm$ 4.6	75.0 $\pm$ 3.4	36.7 $\pm$ 4.3
R3M (CoRL'22)	26.7 $\pm$ 2.3	9.3 $\pm$ 2.3	32.0 $\pm$ 4.0	12.0 $\pm$ 0.0	26.7 $\pm$ 2.3	22.7 $\pm$ 2.3	24.0 $\pm$ 6.9	21.0 $\pm$ 2.6	51.3 $\pm$ 6.0	<u>76.7</u> $\pm$ 2.9	70.0 $\pm$ 2.1	51.7 $\pm$ 6.4	56.3 $\pm$ 4.1
VC-1 (NeurIPS'23)	91.7 $\pm$ 3.5	53.0 $\pm$ 8.7	73.3 $\pm$ 3.8	24.7 $\pm$ 0.6	25.7 $\pm$ 2.1	24.7 $\pm$ 2.1	70.3 $\pm$ 9.6	34.7 $\pm$ 2.3	10.0 $\pm$ 8.7	40.7 $\pm$ 8.1	58.3 $\pm$ 4.4	83.3 $\pm$ 5.8	38.3 $\pm$ 3.3
HRP (CoRL'24)	96.3 $\pm$ 2.9	57.0 $\pm$ 9.2	63.0 $\pm$ 7.2	19.0 $\pm$ 2.0	37.7 $\pm$ 2.3	45.7 $\pm$ 2.1	62.3 $\pm$ 12.7	41.7 $\pm$ 1.2	62.0 $\pm$ 5.0	29.7 $\pm$ 6.4	65.0 $\pm$ 4.8	<u>98.3</u> $\pm$ 2.6	38.3 $\pm$ 3.5
MPI (RSS'24)	<u>98.0</u> $\pm$ 1.7	40.7 $\pm$ 2.9	47.0 $\pm$ 13.1	17.3 $\pm$ 2.6	60.0 $\pm$ 3.7	<u>81.7</u> $\pm$ 8.3	68.0 $\pm$ 9.0	<u>75.0</u> $\pm$ 1.7	55.5 $\pm$ 2.0	71.7 $\pm$ 1.4	66.7 $\pm$ 3.9	53.7 $\pm$ 5.8	43.7 $\pm$ 3.1
MCR (ICLR'25)	75.7 $\pm$ 3.5	50.3 $\pm$ 9.6	42.7 $\pm$ 4.6	20.0 $\pm$ 0.0	28.0 $\pm$ 3.0	26.3 $\pm$ 2.1	34.0 $\pm$ 2.6	24.7 $\pm$ 4.0	59.0 $\pm$ 1.7	71.3 $\pm$ 4.0	60.0 $\pm$ 4.2	46.7 $\pm$ 5.1	36.7 $\pm$ 2.7
R3M-A (CVPR'25)	<u>98.0</u> $\pm$ 0.0	44.0 $\pm$ 6.9	32.3 $\pm$ 4.5	21.7 $\pm$ 2.9	31.0 $\pm$ 5.3	56.7 $\pm$ 6.0	50.3 $\pm$ 2.1	18.3 $\pm$ 2.1	63.3 $\pm$ 2.9	72.7 $\pm$ 0.6	68.3 $\pm$ 3.6	100.0 $\pm$ 0.0	45.0 $\pm$ 2.4
DINOV3 (arXiv'25)	100.0 $\pm$ 0.0	90.7 $\pm$ 2.1	<u>96.3</u> $\pm$ 0.9	<u>75.7</u> $\pm$ 3.4	80.0 $\pm$ 5.0	71.7 $\pm$ 5.7	<u>85.0</u> $\pm$ 4.3	60.3 $\pm$ 5.9	26.0 $\pm$ 8.5	33.3 $\pm$ 10.3	70.0 $\pm$ 4.0	100.0 $\pm$ 0.0	43.3 $\pm$ 3.3
VIP* (ICML'25)	96.0 $\pm$ 3.6	<u>91.0</u> $\pm$ 6.6	63.2 $\pm$ 5.0	68.0 $\pm$ 4.6	56.0 $\pm$ 1.0	73.7 $\pm$ 3.5	79.7 $\pm$ 3.5	71.0 $\pm$ 9.2	30.0 $\pm$ 1.7	64.4 $\pm$ 9.2	73.7 $\pm$ 3.2	96.3 $\pm$ 2.7	55.0 $\pm$ 2.6
LSA	100.0 $\pm$ 0.0	93.3 $\pm$ 2.3	100.0 $\pm$ 0.0	85.3 $\pm$ 2.3	98.0 $\pm$ 2.0	84.0 $\pm$ 0.0	92.3 $\pm$ 1.5	82.7 $\pm$ 1.5	63.9 $\pm$ 3.5	78.3 $\pm$ 1.6	85.0 $\pm$ 1.7	100.0 $\pm$ 0.0	58.3 $\pm$ 1.3

Table 2: Robustness under visual perturbations.

Method	Perturb.	SR(%)	Drop
DINOV3 [44]	Bright.	80.8 $\rightarrow$ 70.8	$\downarrow$ 12.4%
	Disturb.	80.8 $\rightarrow$ 55.0	$\downarrow$ 31.9%
R3M-A [36]	Bright.	43.2 $\rightarrow$ 20.6	$\downarrow$ 52.3%
	Disturb.	43.2 $\rightarrow$ 18.7	$\downarrow$ 56.7%
HRP [8]	Bright.	46.0 $\rightarrow$ 13.8	$\downarrow$ 70.0%
	Disturb.	46.0 $\rightarrow$ 31.0	$\downarrow$ 32.6%
LSA	Bright.	92.2 $\rightarrow$ 85.3	$\downarrow$ 7.5%
	Disturb.	92.2 $\rightarrow$ 75.0	$\downarrow$ 18.7%

Table 3: Ablation study of LSA’s architectural components. **MP**: motion planning; **RA**: representation alignment; **SG**: structural guidance; **AG**: attentive guidance.

	MP	RA	SG	AG	SR	Gain
Exp.1	✓	✓	✓	✓	73.3	+0
Exp.2	✓	✓	✓	✓	81.0	+7.7
Exp.3	✓	✓	✓	✓	85.2	+11.9
Exp.4	✓	✓	✓	✓	85.8	+12.5
Exp.5	✓	✓	✓	✓	86.3	+13.0
Exp.6	✓	✓	✓	✓	89.0	+15.7
Exp.7	✓	✓	✓	✓	88.4	+15.1
Exp.8	✓	✓	✓	✓	92.2	+18.9

238 colored circular occlusions during testing. Tab. 2 shows that LSA suffers smaller drops under these
239 visual perturbations, decreasing by only 7.5% under brightness reduction and 18.7% under visual
240 disturbance. In contrast, the compared baselines show much larger degradation, indicating that LSA
241 is more robust to superficial appearance changes.

242 4.3 Ablation Results

243 Ablation studies are conducted on four MetaWorld tasks, including *dial turn*, *hand insert*, *pick place*
244 *wall*, and *disassemble*. We report the average evaluation success rate across these tasks.

245 **Architecture Design.** As shown in Table 3, the model without human-video pretraining achieves an
246 average success rate of 73.3%. Adding the motion planning objective improves the success rate to
247 81.0%, indicating that future 3D hand trajectories provide transferable action-relevant supervision.
248 Built upon this objective, each component in LSA further improves the learned representation:
249 representation alignment, structural guidance, and attentive guidance individually improve the success
250 rate to 85.2%, 85.8%, and 86.3%, respectively. Combining representation alignment with structural
251 guidance further increases the success rate to 89.0%, suggesting that depth-aware reconstruction
252 further strengthens the dense 3D features learned from representation alignment. Finally, the complete
253 LSA framework achieves the best success rate of 92.2%, improving the non-pretrained baseline
254 by 18.9%. These results show that the three phases of LSA provide complementary supervision,
255 empowering the policy with 3D-aware perception and interaction-aware motion planning abilities
256 learned from human videos.

257 **Analysis of LSA Variants.** Tab. 4 analyzes key design choices in LSA. For motion supervision,
258 using only 2D hand motion achieves 77.3%, slightly below the 3D motion planning baseline of
259 81.0%. However, when the model receives 3D geometric supervision in the Lift and See phases

Table 4: Ablation study on key LSA variants.

Configuration	SR(%)
Full Model (Ours)	92.2
<i>Motion Supervision</i>	
Only 3D Hand Motion	81.0
Only 2D Hand Motion	77.3
3D $\rightarrow$ 2D Hand Motion	56.3 ($\downarrow$ 35.9)
<i>Aligned Representation</i>	
Only VGGT	89.8 ($\downarrow$ 2.4)
Only Depth Pro	90.3 ($\downarrow$ 1.9)
<i>Alignment Weight</i>	
None	89.3 ($\downarrow$ 2.9)
Condition only on c_l	91.2 ($\downarrow$ 1.0)
Condition only on c_m	90.1 ($\downarrow$ 2.1)
<i>Attentive Guidance</i>	
Target: Each head	86.3 ($\downarrow$ 5.9)
Super.: Random head	83.5 ($\downarrow$ 8.7)
Super.: Worst-matching head	81.8 ($\downarrow$ 10.4)
Super.: Hand heatmap	84.3 ($\downarrow$ 7.9)

Table 5: Quality analysis of 3D hand pseudo-labels. ADE denotes the average displacement error in meters.

(a) Wrist 3D lifting accuracy.			
Depth	Intrinsics	H2O ADE $\downarrow$	DexYCB ADE $\downarrow$
GT	GT	0.0000	0.0000
GT	VGGT	0.0093	0.0079
Depth Pro	GT	0.0306	0.0333
Depth Pro	VGGT	0.0393	0.0364
VGGT	GT	0.0974	0.1147
VGGT	VGGT	0.0966	0.1155
(b) Effect of trajectory supervision quality.			
Trajectory Supervision		SR(%)	
VGGT-lifted trajectory		88.6 ($\downarrow$ 3.6)	
Depth Pro + VGGT (Ours)		92.2	
DexYCB GT trajectory		93.5 ($\uparrow$ 1.3)	

but uses 2D trajectory prediction in the Act phase, the performance drops further. We attribute this to the semantic inconsistency between the lifted 3D representation space and the image-plane 2D motion objective. Using only VGGT or Depth Pro for representation alignment is inferior to using both, indicating that different 3D FMs provide complementary dense geometric cues. Removing dynamic alignment weights or conditioning them on a single layer also degrades performance, showing the need for adaptive multi-teacher weighting with multi-layer geometric context. For attentive guidance, supervising the attention of all shallow heads suppresses attention diversity and thus degrades performance. In addition, using a random head, the worst-matching head, or the hand heatmap as the supervision target also reduces performance, indicating that the internally emerged attention target selected by the hand heatmap more effectively captures interaction regions.

Quality of 3D Pseudo-labels. Tab. 5 evaluates the reliability of the pseudo-3D hand labels used by LSA. To analyze the error, we separately evaluate lifting accuracy on H2O and DexYCB, using 240 valid wrist samples from 12 scenes for each dataset; more details are provided in **Appendix B.3**. Depth Pro depth with VGGT intrinsics substantially reduces wrist lifting errors compared with VGGT depth variants, e.g., 0.0364 m vs. 0.1155 m on DexYCB. When replacing only the Act-phase trajectory labels, noisier VGGT-lifted trajectories reduce SR from 92.2% to 88.6%, while DexYCB GT trajectories only improve SR to 93.5%. The limited improvement from replacing our pseudo-trajectories with GT trajectories suggests that the adopted Depth Pro+VGGT pipeline provides reasonably reliable supervision in our setting. We also validate the effectiveness of our labeling and pretraining strategy on more diverse unlabeled datasets, such as Ego4D (**Appendix C.3**).

More Analyses. Given the limited space, we provide additional analyses in the **Appendix**, including detailed experimental protocols (**Appendix B**), data scaling and diversity studies (**Appendix C.1, C.3**), parameter update strategies (**Appendix C.2**), hyperparameter and layer-selection ablations (**Appendix C.4, C.5**), training convergence (**Appendix C.6**), and more visualizations of attentive guidance and task trajectories (**Appendix D**).

5 Real-World Experiments

5.1 Setup

We further evaluate LSA in real-world manipulation using a 7-DoF Franka Research 3 robot equipped with a 1-DoF parallel gripper. RGB observations are captured by a RealSense D435i camera. As shown in Fig. 4, we consider four daily-scenario manipulation tasks: *Wipe PC*, *Pick Trash*, *Place Toy*, and *Pluck Flower*. For each task, we collect 30 demonstrations with variations in the initial poses of the robot and objects. We evaluate each policy over ten trials per task and report the success rate (%).

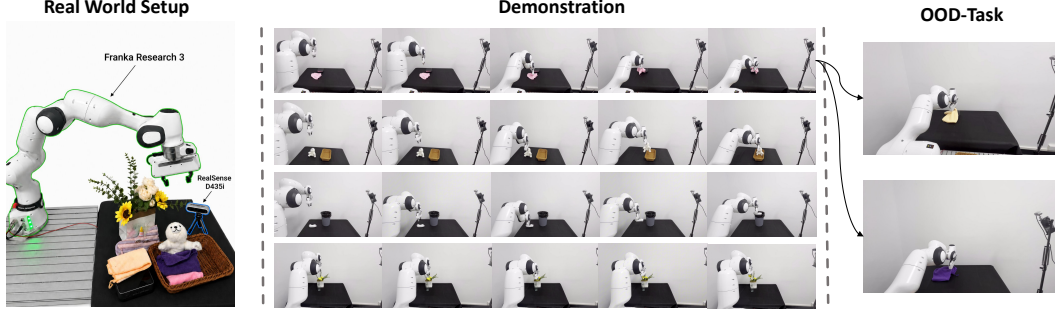


Figure 4: Overview of our real-world experiment setup. The figure shows our robotic platform and sensors (left), the execution of the four tasks under the in-distribution setup (middle), and the modifications applied for the out-of-distribution setup (right).

Table 6: Real-world evaluation with the Franka robot across four tasks. For *Wipe PC*, *U1* and *U2* denote two unseen towel colors.

Method	Pluck Flower	Pick Trash	Place Toy	Wipe PC	Wipe PC U1	Wipe PC U2	Avg.
DINOv3	20.0	30.0	20.0	30.0	10.0	20.0	21.7
R3M-A	10.0	30.0	30.0	40.0	0.0	10.0	20.0
VIP	50.0	40.0	50.0	40.0	20.0	20.0	36.7
HRP	40.0	40.0	60.0	50.0	20.0	40.0	41.7
Ours	50.0	70.0	80.0	80.0	70.0	60.0	68.3

To evaluate out-of-distribution (OOD) generalization, we further construct two unseen variants of *Wipe PC*. The policy is trained with one towel color and tested with two previously unseen towel colors, denoted as *unseen1* and *unseen2*. This setting tests whether the learned representation can generalize beyond superficial appearance changes. More implementation details are provided in Appendix B.4.

5.2 Evaluation Results

Tab. 6 shows that LSA achieves the highest average success rate among all compared methods, with 68.3% vs. 41.7% for the strongest baseline HRP. LSA outperforms the second-best method on *Pick Trash* (70.0% vs. 40.0%), *Place Toy* (80.0% vs. 60.0%), and *Wipe PC* (80.0% vs. 50.0%), while matching the best result on *Pluck Flower* (50.0%). On the two OOD *Wipe PC* settings, LSA obtains 70.0% and 60.0%, whereas the best baselines achieve 20.0% and 40.0%; notably, R3M-A drops to 0.0% on *Wipe PC U1*. These results suggest that LSA transfers better to real-world manipulation. On one hand, its human-video pretraining learns 3D-aware representations from 2D observations, reducing the burden of acquiring spatial relations from limited robot demonstrations. On the other hand, its interaction-focused guidance helps the policy preserve task-relevant hand-object relations under appearance changes.

6 Conclusion

In this paper, we propose LSA, a hierarchical robot policy pretraining framework that systematically decomposes the learning process into three synergistic phases: Lift, See, and Act. Unlike prior works that rely solely on sparse 3D trajectories, LSA fully exploits the multi-tiered knowledge of 3D FMs by aligning policy features with dense, intermediate geometric representations. To bridge the gap between implicit features and physical execution, we introduce a dual-guidance mechanism in the See phase, combining explicit structural reconstruction with attentive guidance to focus on task-relevant regions. Finally, the Act phase grounds the learned representation in future 3D hand motion, which serves as an action-relevant proxy for downstream robot manipulation. **Limitations.** LSA shares several limitations inherent to human-video-based robot pretraining methods due to the human-robot embodiment gap. **Broader Impacts.** Looking forward, this work opens avenues for scaling multi-tiered 3D knowledge extraction from human videos to broader robot manipulation scenarios, with the potential to further enhance Vision-Language-Action model pretraining. To the best of our knowledge, this work has no obvious negative social impacts.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [2] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.
- [3] Chubin Zhang, Jianan Wang, Zifeng Gao, Yue Su, Tianru Dai, Cai Zhou, Jiwen Lu, and Yansong Tang. Clap: Contrastive latent action pretraining for learning vision-language-action models from human videos. *arXiv preprint arXiv:2601.04061*, 2026.
- [4] Shenyan Gao, William Liang, Kaiyuan Zheng, Ayaan Malik, Seonghyeon Ye, Sihyun Yu, Wei-Cheng Tseng, Yuzhu Dong, Kaichun Mo, Chen-Hsuan Lin, et al. Dreamdojo: A generalist robot world model from large-scale human videos. *arXiv preprint arXiv:2602.06949*, 2026.
- [5] Zhiyuan Feng, Qixiu Li, Huizhi Liang, Rushuai Yang, Yichao Shen, Zhiying Du, Zhaowei Zhang, Yu Deng, Li Zhao, Hao Zhao, et al. From human videos to robot manipulation: A survey on scalable vision-language-action learning with human-centric data. *Authorea Preprints*, 2026.
- [6] Alexey Gavryushin, Xi Wang, Robert JS Malate, Chenyu Yang, Xiangyi Jia, Shubh Goel, Davide Liconti, René Zurbrugg, Robert K Katzschmann, and Marc Pollefeys. Maple: Encoding dexterous robotic manipulation priors learned from egocentric videos. *arXiv preprint arXiv:2504.06084*, 2025.
- [7] Shikhar Bahl, Russell Mendonca, Lili Chen, Unnat Jain, and Deepak Pathak. Affordances from human videos as a versatile representation for robotics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13778–13790, 2023.
- [8] Mohan Kumar Srirama, Sudeep Dasari, Shikhar Bahl, and Abhinav Gupta. Hrp: Human affordances for robotic pre-training. In *Robotics: Science and Systems*, 2024.
- [9] Jia Zeng, Qingwen Bu, Bangjun Wang, Wenke Xia, Li Chen, Hao Dong, Haoming Song, Dong Wang, Di Hu, Ping Luo, et al. Learning manipulation by predicting interaction. In *Robotics: Science and Systems*, 2024.
- [10] Hao Luo, Yicheng Feng, Wanpeng Zhang, Sipeng Zheng, Ye Wang, Haoqi Yuan, Jiazheng Liu, Chaoyi Xu, Qin Jin, and Zongqing Lu. Being-h0: Vision-language-action pretraining from large-scale human videos. *arXiv preprint arXiv:2507.15597*, 2025.
- [11] Hanzhi Chen, Boyang Sun, Anran Zhang, Marc Pollefeys, and Stefan Leutenegger. Vidbot: Learning generalizable 3d actions from in-the-wild 2d human videos for zero-shot robotic manipulation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 27661–27672, 2025.
- [12] Qixiu Li, Yu Deng, Yaobo Liang, Lin Luo, Lei Zhou, Chengtang Yao, Lingqi Zeng, Zhiyuan Feng, Huizhi Liang, Sicheng Xu, et al. Scalable vision-language-action model pretraining for robotic manipulation with real-life human activity videos. *arXiv preprint arXiv:2510.21571*, 2025.
- [13] Juncheng Mu, Sizhe Yang, Yiming Bao, Hojin Bae, Tianming Wei, Linning Xu, Boyi Li, Huazhe Xu, and Jiangmiao Pang. Deximit: Learning bimanual dexterous manipulation from monocular human videos. *arXiv preprint arXiv:2602.10105*, 2026.
- [14] Hongyi Chen, Tony Dong, Tiancheng Wu, Liquan Wang, Yash Jangir, Yaru Niu, Yufei Ye, Homanga Bharadhwaj, Zackory Erickson, and Jeffrey Ichnowski. Dexterous manipulation policies from rgb human videos via 3d hand-object trajectory reconstruction. *arXiv preprint arXiv:2602.09013*, 2026.
- [15] Jianyuan Wang, Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.
- [16] Aleksei Bochkovskii, Amaël Delaunoy, Hugo Germain, Marcel Santos, Yichao Zhou, Stephan R Richter, and Vladlen Koltun. Depth pro: Sharp monocular metric depth in less than a second. *arXiv preprint arXiv:2410.02073*, 2024.
- [17] Ruicheng Wang, Sicheng Xu, Yue Dong, Yu Deng, Jianfeng Xiang, Zelong Lv, Guangzhong Sun, Xin Tong, and Jiaolong Yang. Moge-2: Accurate monocular geometry with metric scale and sharp details. *arXiv preprint arXiv:2507.02546*, 2025.

- [18] Jinglei Zhang, Jiankang Deng, Chao Ma, and Rolandos Alexandros Potamias. Hawor: World-space hand motion reconstruction from egocentric videos. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 1805–1815, 2025.
- [19] Suraj Nair, Aravind Rajeswaran, Vikash Kumar, Chelsea Finn, and Abhinav Gupta. R3m: A universal visual representation for robot manipulation. *arXiv preprint arXiv:2203.12601*, 2022.
- [20] Yecheng Jason Ma, Shagun Sodhani, Dinesh Jayaraman, Osbert Bastani, Vikash Kumar, and Amy Zhang. Vip: Towards universal visual reward and representation via value-implicit pre-training. *arXiv preprint arXiv:2210.00030*, 2022.
- [21] Aaron Van Den Oord, Oriol Vinyals, et al. Neural discrete representation learning. *Advances in neural information processing systems*, 30, 2017.
- [22] Seonghyeon Ye, Joel Jang, Byeongguk Jeon, Sejeun Joo, Jianwei Yang, Baolin Peng, Ajay Mandlekar, Reuben Tan, Yu-Wei Chao, Bill Yuchen Lin, et al. Latent action pretraining from videos. *arXiv preprint arXiv:2410.11758*, 2024.
- [23] Jiange Yang, Yansong Shi, Haoyi Zhu, Mingyu Liu, Kaijing Ma, Yating Wang, Gangshan Wu, Tong He, and Limin Wang. Como: Learning continuous latent motion from internet videos for scalable robot learning. *arXiv preprint arXiv:2505.17006*, 2025.
- [24] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. In *International Conference on Machine Learning*, pages 24328–24346. PMLR, 2025.
- [25] Homanga Bharadhwaj, Debidatta Dwibedi, Abhinav Gupta, Shubham Tulsiani, Carl Doersch, Ted Xiao, Dhruv Shah, Fei Xia, Dorsa Sadigh, and Sean Kirmeni. Gen2act: Human video generation in novel scenarios enables generalizable robot manipulation. *arXiv preprint arXiv:2409.16283*, 2024.
- [26] Sandeep Routray, Hengkai Pan, Unnat Jain, Shikhar Bahl, and Deepak Pathak. Vipra: Video prediction for robot actions. *arXiv preprint arXiv:2511.07732*, 2025.
- [27] Gen Li, Nikolaos Tsagkas, Jifei Song, Ruaridh Mon-Williams, Sethu Vijayakumar, Kun Shao, and Laura Sevilla-Lara. Learning precise affordances from egocentric videos for robotic manipulation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10581–10591, 2025.
- [28] Marion Lepert, Jiaying Fang, and Jeannette Bohg. Masquerade: Learning from in-the-wild human videos using data-editing. *arXiv preprint arXiv:2508.09976*, 2025.
- [29] Javier Romero, Dimitrios Tzionas, and Michael J Black. Embodied hands: Modeling and capturing hands and bodies together. *arXiv preprint arXiv:2201.02610*, 2022.
- [30] Ruihan Yang, Qinxu Yu, Yecheng Wu, Rui Yan, Borui Li, An-Chieh Cheng, Xueyan Zou, Yunhao Fang, Xuxin Cheng, Ri-Zhao Qiu, et al. Egovla: Learning vision-language-action models from egocentric human videos. *arXiv preprint arXiv:2507.12440*, 2025.
- [31] Angela Dai, Matthias Nießner, Michael Zollhöfer, Shahram Izadi, and Christian Theobalt. Bundlefusion: Real-time globally consistent 3d reconstruction using on-the-fly surface reintegration. *ACM Transactions on Graphics (ToG)*, 36(4):1, 2017.
- [32] Hongzhe Bi, Lingxuan Wu, Tianwei Lin, Hengkai Tan, Zhizhong Su, Hang Su, and Jun Zhu. H-rdt: Human manipulation enhanced bimanual robotic manipulation. *arXiv preprint arXiv:2507.23523*, 2025.
- [33] Xiongqi Cai, Ri-Zhao Qiu, Geng Chen, Lai Wei, Isabella Liu, Tianshu Huang, Xuxin Cheng, and Xiaolong Wang. In-n-on: Scaling egocentric manipulation with in-the-wild and on-task data. *arXiv preprint arXiv:2511.15704*, 2025.
- [34] Tomoya Yoshida, Shuhei Kurita, Taichi Nishimura, and Shinsuke Mori. Developing vision-language-action model from egocentric videos. *arXiv preprint arXiv:2509.21986*, 2025.
- [35] Younggyo Seo, Danijar Hafner, Hao Liu, Fangchen Liu, Stephen James, Kimin Lee, and Pieter Abbeel. Masked world models for visual control. In *Conference on Robot Learning*, pages 1332–1344. PMLR, 2023.
- [36] Jiaming Zhou, Teli Ma, Kun-Yu Lin, Zifan Wang, Ronghe Qiu, and Junwei Liang. Mitigating the human-robot domain discrepancy in visual pre-training for robotic manipulation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 22551–22561, 2025.

- [37] Michael Bain and Claude Sammut. A framework for behavioural cloning. In *Machine intelligence 15*, pages 103–129, 1995.
- [38] Sihyun Yu, Sangkyung Kwak, Huiwon Jang, Jongheon Jeong, Jonathan Huang, Jinwoo Shin, and Saining Xie. Representation alignment for generation: Training diffusion transformers is easier than you think. *arXiv preprint arXiv:2410.06940*, 2024.
- [39] Jiajun Cao, Yuan Zhang, Tao Huang, Ming Lu, Qizhe Zhang, Ruichuan An, Ningning Ma, and Shanghang Zhang. Move-kd: Knowledge distillation for vlms with mixture of visual encoders. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 19846–19856, 2025.
- [40] Tianhe Yu, Deirdre Quillen, Zhanpeng He, Ryan Julian, Karol Hausman, Chelsea Finn, and Sergey Levine. Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *Conference on robot learning*, pages 1094–1100. PMLR, 2020.
- [41] Chen Bao, Helin Xu, Yuzhe Qin, and Xiaolong Wang. Dexart: Benchmarking generalizable dexterous manipulation with articulated objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 21190–21200, 2023.
- [42] Aravind Rajeswaran, Vikash Kumar, Abhishek Gupta, Giulia Vezzani, John Schulman, Emanuel Todorov, and Sergey Levine. Learning complex dexterous manipulation with deep reinforcement learning and demonstrations. *arXiv preprint arXiv:1709.10087*, 2017.
- [43] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021.
- [44] Oriane Siméoni, Huy V Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. Dinov3. *arXiv preprint arXiv:2508.10104*, 2025.
- [45] Guangqi Jiang, Yifei Sun, Tao Huang, Huanyu Li, Yongyuan Liang, and Huazhe Xu. Robots pre-train robots: Manipulation-centric robotic representation from large-scale robot datasets. *arXiv preprint arXiv:2410.22325*, 2024.
- [46] Arjun Majumdar, Karmesh Yadav, Sergio Arnaud, Jason Ma, Claire Chen, Sneha Silwal, Aryan Jain, Vincent-Pierre Berges, Tingfan Wu, Jay Vakil, et al. Where are we in the search for an artificial visual cortex for embodied intelligence? *Advances in Neural Information Processing Systems*, 36:655–677, 2023.
- [47] Yanjie Ze, Gu Zhang, Kangning Zhang, Chenyuan Hu, Muhan Wang, and Huazhe Xu. 3d diffusion policy: Generalizable visuomotor policy learning via simple 3d representations. In *Proceedings of Robotics: Science and Systems (RSS)*, 2024.
- [48] Zhihao Chen, Yiyuan Ge, and Ziyang Wang. Kan we flow? advancing robotic manipulation with 3d flow matching via kan & rwkv. *arXiv preprint arXiv:2602.01115*, 2026.
- [49] Fan Zhang, Valentin Bazarevsky, Andrey Vakunov, Andrei Tkachenka, George Sung, Chuo-Ling Chang, and Matthias Grundmann. Mediapipe hands: On-device real-time hand tracking. *arXiv preprint arXiv:2006.10214*, 2020.
- [50] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- [51] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255. Ieee, 2009.
- [52] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.
- [53] Himanshu Gaurav Singh, Antonio Loquercio, Carmelo Sferrazza, Jane Wu, Haozhi Qi, Pieter Abbeel, and Jitendra Malik. Hand-object interaction pretraining from videos. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 3352–3360. IEEE, 2025.
- [54] Yu-Wei Chao, Wei Yang, Yu Xiang, Pavlo Molchanov, Ankur Handa, Jonathan Tremblay, Yashraj S Narang, Karl Van Wyk, Umar Iqbal, Stan Birchfield, et al. Dexycb: A benchmark for capturing hand grasping of objects. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9044–9053, 2021.

- 474 [55] Yingdong Hu, Renhao Wang, Li Erran Li, and Yang Gao. For pre-trained vision models in motor control,
475 not all policy learning methods are created equal. In *International Conference on Machine Learning*, pages
476 13628–13651. PMLR, 2023.
- 477 [56] Nikolaos Tsagkas, Andreas Sochopoulos, Duolikun Danier, Chris Xiaoxuan Lu, and Oisín Mac Aodha.
478 When pre-trained visual representations fall short: Limitations in visuo-motor robot learning. *arXiv*
479 *e-prints*, pages arXiv–2502, 2025.
- 480 [57] Emanuel Todorov, Tom Erez, and Yuval Tassa. Mujoco: A physics engine for model-based control. In
481 *2012 IEEE/RSJ international conference on intelligent robots and systems*, pages 5026–5033. IEEE, 2012.
- 482 [58] Che Wang, Xufang Luo, Keith Ross, and Dongsheng Li. Vrl3: A data-driven framework for visual deep
483 reinforcement learning. *Advances in Neural Information Processing Systems*, 35:32974–32988, 2022.
- 484 [59] Taein Kwon, Bugra Tekin, Jan Stühmer, Federica Bogo, and Marc Pollefeys. H2o: Two hands manipulating
485 objects for first person interaction recognition. In *Proceedings of the IEEE/CVF international conference*
486 *on computer vision*, pages 10138–10148, 2021.
- 487 [60] Wentao Bao, Lele Chen, Libing Zeng, Zhong Li, Yi Xu, Junsong Yuan, and Yu Kong. Uncertainty-aware
488 state space transformer for egocentric 3d hand trajectory forecasting. In *Proceedings of the IEEE/CVF*
489 *international conference on computer vision*, pages 13702–13711, 2023.
- 490 [61] Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar,
491 Jackson Hamburger, Hao Jiang, Miao Liu, Xingyu Liu, et al. Ego4d: Around the world in 3,000 hours of
492 egocentric video. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*,
493 pages 18995–19012, 2022.

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A Additional Methodology and Implementation Details

This section provides additional details of the LSA pretraining framework, including the overall pretraining objective, dynamic weight allocation, attentive guidance, data cleaning, 3D lifting, hand prior construction, and pretraining implementation.

A.1 Overall Pretraining Objective

The overall supervision loss consists of four terms: the motion planning loss $\mathcal{L}_{\text{motion}}$, the representation alignment loss $\mathcal{L}_{\text{align}}$, the depth reconstruction loss $\mathcal{L}_{\text{recon}}$, and the attentive guidance loss $\mathcal{L}_{\text{att}}$. The total loss in the pretraining stage is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{motion}} + \lambda_g \cdot (\mathcal{L}_{\text{align}} + \mathcal{L}_{\text{recon}} + \lambda_a \cdot \mathcal{L}_{\text{att}}), \quad (5)$$

where λ_g balances the geometry-related supervision terms and λ_a controls the strength of attentive guidance. In our experiments, we set $\lambda_g = 0.3$ and $\lambda_a = 0.5$.

A.2 Dynamic Weight Allocation

We design the dynamic weight assignment mechanism to efficiently harness representations from multiple 3D FMs, as the optimal contribution of each FM fundamentally varies across diverse manipulation scenarios. To adaptively learn the most valuable knowledge, we formulate a condition query leveraging the [CLS] tokens c_l and c_m from the representation alignment layer (l -th layer) and the depth reconstruction layer (m -th layer), respectively. As summarized in **Algorithm 1**, this condition query is used to compute the scaled dot-product similarity with geometric tokens and produce the dynamic teacher weights. First, the [CLS] token naturally serves as a compact global representation, effectively encapsulating the spatial semantics of the entire scene. Second, the two selected [CLS] tokens capture geometry-related semantics. Specifically, the l -th layer is directly aligned with the implicit intermediate representations of the 3D FMs, thus capturing **implicit geometric information**. Conversely, the m -th layer is explicitly optimized for the depth reconstruction auxiliary task, thereby encoding the **explicit 3D spatial geometry** of the physical world. By aggregating them, we formulate a condition query that achieves a comprehensive geometric awareness, thereby better modulating the teacher weights.

Algorithm 1 Dynamic Weight Allocation

Input: $\mathbf{c}_l, \mathbf{c}_m$ from l -th and m -th layers; geometric tokens $\mathbf{E}'_c, \mathbf{E}'_d$.

Output: Global modulation weights w_c, w_d .

```

1:  $\mathbf{Q} \leftarrow \text{MeanPool}(\text{Concat}(\mathbf{c}_l, \mathbf{c}_m)) \in \mathbb{R}^{B \times D}$  {Query}
2: for all  $j \in \{c, d\}$  do
3:    $\hat{\mathbf{E}}'_j \leftarrow \text{MeanPool}(\mathbf{E}'_j) \in \mathbb{R}^{B \times N_a \times D}$  {Projection}
4:    $\mathbf{S}_j \leftarrow (\mathbf{Q} \hat{\mathbf{E}}'^{\top}_j) / \sqrt{D}$ 
5:    $\mathbf{s}_j \leftarrow \frac{1}{N_a} \sum_{i=1}^{N_a} \mathbf{S}_{j,i}$  {Similarity}
6: end for
7:  $\mathbf{G} \leftarrow \text{Softmax}([\mathbf{s}_c, \mathbf{s}_d]) \in \mathbb{R}^{B \times 2}$  {Gating}
8:  $w_c \leftarrow \text{avg}(\mathbf{G}_{:,0}); \quad w_d \leftarrow \text{avg}(\mathbf{G}_{:,1})$  {Aggregation}
9: return  $w_c, w_d$ 

```

520 A.3 Attentive Guidance Details

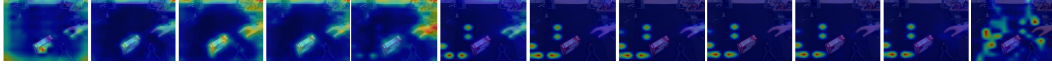


Figure 5: Attention maps of the [CLS] token from layers 1 to 12 in LSA without attentive guidance. Shallow layers tend to focus on local regions such as hands and objects, while deeper layers become more dispersed and abstract.

521 We further provide details and visual evidence for attentive guidance. We treat the [CLS] tokens from
 522 different ViT layers as distinct entities, since they encode hierarchical global semantics. As shown
 523 in Fig. 5, the attention maps associated with these layer-specific [CLS] tokens also evolve across
 524 layers. Shallow layers tend to focus on local regions such as hands and objects, whereas deeper layers
 525 become more abstract and dispersed.

526 Based on this observation, attentive guidance uses the hand region only as a cue to mine an internal
 527 task-relevant attention map. As shown in Fig. 6, the original ViT produces relatively dispersed
 528 shallow attention, while motion planning encourages clearer attention to the hand region and further
 529 induces emergent attention to the interacting object. This suggests that future hand trajectories
 530 provide supervision signals that implicitly capture hand-object spatial relations. We therefore select
 531 an internal attention map that covers both the hand and the interacting object using the hand prior.
 532 Compared with supervising each individual head, our method guides only the head-averaged attention
 533 map of each shallow layer, aligning the overall spatial focus while preserving inter-head diversity.

534 A.4 Data Cleaning and 3D Lifting

535 We use DexYCB as the main pretraining data source. Our data cleaning process consists of three
 536 steps.

537 First, we detect the hand pose in each frame. Due to occlusions and detection failures, we remove a
 538 video if the hand is not detected for ≥ 3 consecutive frames.

539 Second, given a per-frame metric depth sequence $\{z_t\}_{t=1}^T$ in a video, we compute its temporal
 540 standard deviation $\sigma_z = \text{std}(\{z_t\})$ and frame-to-frame depth differences $\Delta z_t = |z_t - z_{t-1}|$. A video
 541 is discarded as low quality if it is globally unstable, i.e., $\sigma_z > 0.09$ and $\frac{1}{T-1} \sum_{t=2}^T \Delta z_t > 0.02$, or if
 542 it contains an abrupt jump, i.e., $\max_t \Delta z_t > 0.15$. These thresholds are set empirically to remove
 543 unstable lifted trajectories.

544 Third, we apply Gaussian smoothing to the metric depth sequence $\{z_t\}$ with $\sigma = 3.0$ and obtain the
 545 smoothed depth sequence $\{\tilde{z}_t\}$. The first and last depth values are kept unchanged to avoid boundary
 546 artifacts.

547 After data cleaning, we lift each 2D wrist keypoint (u_t, v_t) into 3D space using the smoothed metric
 548 depth $\tilde{z}_t$ and camera intrinsics. Let (f_x, f_y) denote the focal lengths and (c_x, c_y) denote the principal
 549 point. The lifted 3D wrist point $\mathbf{p}_t$ in the camera coordinate system is computed by the pinhole



Figure 6: Comparison of attention maps. **Left:** averaged attention map over the first $R = 6$ layers of the original ViT. **Middle:** averaged attention map after motion-planning pretraining. **Right:** target attention map selected by the hand prior. The selected target attends to both the hand and the interacting object.

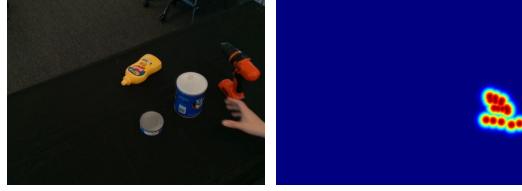


Figure 7: **Left:** Original image. **Right:** Generated hand prior heatmap.

camera model:

$$\mathbf{p}_t = \begin{bmatrix} X_t \\ Y_t \\ Z_t \end{bmatrix} = \begin{bmatrix} (u_t - c_x)\tilde{z}_t/f_x \\ (v_t - c_y)\tilde{z}_t/f_y \\ \tilde{z}_t \end{bmatrix}, \quad t = 1, \dots, T. \quad (6)$$

A.5 Hand Prior Heatmap

To construct the spatial hand prior used in attentive guidance, we employ MediaPipe [49] to extract the 2D image coordinates of 21 hand joints. Let $\mathbf{p}_k = (x_k, y_k)$ denote the 2D coordinate of the k -th hand joint. We place a Gaussian kernel at each detected keypoint and aggregate the responses using the maximum value at each pixel location:

$$H(x, y) = \max_{k \in \{1, \dots, 21\}} \exp \left(-\frac{(x - x_k)^2 + (y - y_k)^2}{2\sigma^2} \right). \quad (7)$$

Here, σ controls the spatial spread of the local response. We further convert H into a softened spatial prior using a temperature-scaled Softmax:

$$\mathbf{h}(x, y) = \frac{\exp(H(x, y)/\tau)}{\sum_{i,j} \exp(H(i, j)/\tau)}. \quad (8)$$

This operation smooths the distribution and allows the hand prior to cover not only exact keypoint locations but also nearby hand regions and interaction context. In our experiments, we set $\sigma = 12$ and $\tau = 5$. Fig. 7 shows the results.

A.6 Pretraining Implementation Details

We adopt a pretrained ViT-B [50] initialized on ImageNet [51] as the visual encoder. To enable parameter-efficient adaptation, we keep the original ViT-B parameters frozen and apply LoRA [52] with rank $r = 8$ and scaling factor $\alpha = 16$. Following our hierarchical design, we perform representation alignment at the $l = 3$ layer of the visual encoder, while the visual tokens from the $m = 6$ layer are routed to an 8-layer Transformer decoder for metric depth reconstruction. Attentive guidance is applied to the first $R = 6$ layers. Attentive guidance is enabled after a 10-epoch warm-up with the motion planning objective, so that the model can first develop initial task-relevant attention patterns before the target attention map is selected. For the Act phase, the model predicts the future 3D wrist trajectory for the next $L = 3$ frames.

Following prior work [53], we use DexYCB [54] to pretrain LSA. After data cleaning, we retain 20K video clips for pretraining. Each clip consists of consecutive frames that depict human hands

573 performing tasks and moving within the scene. Optimization is performed using Adam with a cosine
574 learning rate schedule and an initial learning rate of 5×10^{-4} . The model is trained for 50 epochs on
575 four NVIDIA RTX 4090 GPUs with 24GB memory each, using a batch size of 16 per GPU.

576 B Experimental Protocols

577 This section provides detailed experimental protocols for simulation benchmarks, baseline implemen-
578 tations, 3D wrist lifting evaluation, and real-world experiments.

579 B.1 Simulation Evaluation Setup

580 We provide detailed simulation settings for the three benchmarks used in the main paper: Meta-
581 World [40], DexArt [41], and Adroit [42].

582 **MetaWorld.** MetaWorld provides tabletop manipulation tasks with a Sawyer arm and gripper. We
583 select eight tasks spanning different difficulty levels [35]: regular tasks including *hammer* and *dial*
584 *turn*, hard tasks including *assembly* and *hand insert*, and very hard tasks including *pick place wall*,
585 *shelf place*, *disassemble*, and *stick pull*. For each task, we use scripted policies to collect 25 expert
586 demonstrations. Following prior works [55, 56], the maximum rollout length is strictly limited to
587 175 steps, which makes the setting more challenging by requiring faster and more accurate task
588 completion. During downstream policy learning, the frozen visual encoder extracts visual features
589 from image observations. These visual features are concatenated with the robot proprioceptive state
590 and then fed into the policy head. The behavior cloning policy is trained for 20,000 steps with a batch
591 size of 32 and evaluated every 2,000 steps. **Each evaluation contains 25 independent rollouts.**

592 **DexArt.** DexArt focuses on dexterous manipulation of articulated objects using an XArm6 arm and
593 an Allegro hand. We evaluate on two tasks: *bucket* and *toilet*. For each task, we collect 100 expert
594 demonstrations using pretrained reinforcement learning policies. Each rollout is capped at 100 steps.
595 The behavior cloning policy is trained for 3,000 epochs with a batch size of 64 and evaluated every
596 200 epochs. **Each evaluation contains 20 independent rollouts.** The pretrained visual encoder is
597 frozen during downstream policy learning, and only the policy head is trained.

598 **Adroit.** Adroit simulates a multi-fingered Shadow Hand in MuJoCo [57] and contains contact-rich
599 dexterous manipulation tasks involving rigid and articulated objects. We evaluate on three tasks: *door*,
600 *hammer*, and *pen*. For each task, we collect 10 expert demonstrations using a pretrained reinforcement
601 learning policy [58]. Each demonstration contains 100 steps. The training and evaluation protocols
602 follow the DexArt setting. We include this benchmark to further evaluate whether LSA can transfer
603 to complex dexterous manipulation with fine-grained hand-object interactions.

604 **Reporting Protocol.** For each evaluation checkpoint, we compute the success rate over all rollouts.
605 For each random seed, we average the success rates of the top-3 performing checkpoints. We then
606 report the mean and sample standard deviation over three random seeds, rounded to one decimal
607 place.

608 B.2 Baseline Details

609 We compare LSA with three categories of baselines.

610 **General-purpose Visual Representations.** We include CLIP [43] and DINOv3 [44] as strong
611 general visual representation baselines. CLIP is a vision-language model trained with contrastive
612 image-text supervision, which learns transferable semantic visual features from large-scale image-
613 caption pairs. DINOv3 is a self-supervised vision foundation model that learns strong image-level
614 and dense visual features without manual labels. Although both models provide powerful general-
615 purpose representations, their pretraining objectives are not explicitly designed for action-relevant 3D
616 reasoning or robot manipulation.

617 **Robot Visual Pretraining Methods.** We compare with MCR [45], VIP [20], R3M [19], VC-1 [46],
618 HRP [8], MPI [9], and R3M-A [36]. These methods represent strong robot-oriented visual pretraining
619 baselines, covering different ways of learning action-relevant representations for downstream robotic
620 control.

Direct 3D Policies. We further compare with DP3 [47] and KAN-We-Flow [48], both of which directly take 3D point clouds as policy inputs and learn action prediction through diffusion- or flow-matching-based objectives. These methods provide direct comparisons to LSA from the perspective of 3D-aware policy learning.

For all baselines, we use the same downstream demonstrations as LSA. The pretrained visual encoder is frozen and combined with the same lightweight behavior cloning policy head. The policy head is implemented as a 3-layer MLP with hidden size 256, following prior works [45, 8].

B.3 3D Wrist Lifting Evaluation Set Construction

To quantify the noise of the lifted 3D wrist trajectories, we construct two small evaluation sets from H2O [59] and DexYCB [54], respectively. For each dataset, we randomly sample 240 valid wrist samples from 12 scenes, covering either left or right hands. Both datasets provide calibrated cameras and ground-truth 3D hand poses. For each sample, we use the ground-truth 2D wrist projection as the wrist location, estimate depth and camera intrinsics from different sources, and lift the wrist point to the camera coordinate system. Following [60], we report the 3D Average Displacement Error (ADE) between the lifted wrist and the ground-truth 3D wrist.

B.4 Real-World Implementation Details

Hardware Setup. We conduct real-world experiments using a 7-DoF Franka Research 3 robot equipped with a 1-DoF parallel gripper. A RealSense D435i camera is used to capture RGB observations from a fixed third-person viewpoint. During data collection, we record robot states and RGB observations at 16 Hz.

Demonstration Collection. For each task, we collect 30 teleoperated demonstrations. The demonstrations include variations in the initial poses of the robot and objects to avoid learning a fixed execution trajectory. The four real-world tasks are defined as follows:

- *Wipe PC*: the robot grasps a pink towel and wipes the surface of a laptop.
- *Pick Trash*: the robot picks up a piece of trash and drops it into a trash bin.
- *Place Toy*: the robot grasps a toy and places it into a basket.
- *Pluck Flower*: the robot grasps a flower and lifts it upward.

Policy Training. Following prior real-world policy evaluation protocols, we freeze the pretrained visual encoder and train only the downstream policy head. The policy head is a 3-layer MLP with two hidden layers of size 512. We train the policy using a log-likelihood objective with a learning rate of 1×10^{-4} and a batch size of 128 for 50k steps. The predicted actions are linearly interpolated to 1000 Hz to satisfy the Franka control frequency.

Evaluation Protocol. For each task, we evaluate each policy over 10 independent trials and report the success rate. A trial is considered successful if the robot completes the target manipulation goal without human intervention. For the OOD variants of *Wipe PC*, the policy is trained with a pink towel and evaluated with two unseen towel colors, purple and yellow. This setting changes the object appearance while preserving the same task goal and interaction structure.

C Additional Experimental Results

This section provides additional experimental analyses that are not included in the main paper due to space limitations. Unless otherwise specified, ablation studies are conducted on four MetaWorld tasks, including *dial turn*, *hand insert*, *pick place wall*, and *disassemble*. We report the average evaluation success rate across these tasks.

C.1 Data Scaling and Sample Efficiency

We first investigate the effect of data scale during both human-video pretraining and downstream policy learning. As shown in Fig. 8, increasing the amount of pretraining video data consistently improves downstream success rates, indicating that LSA can benefit from larger-scale human-video

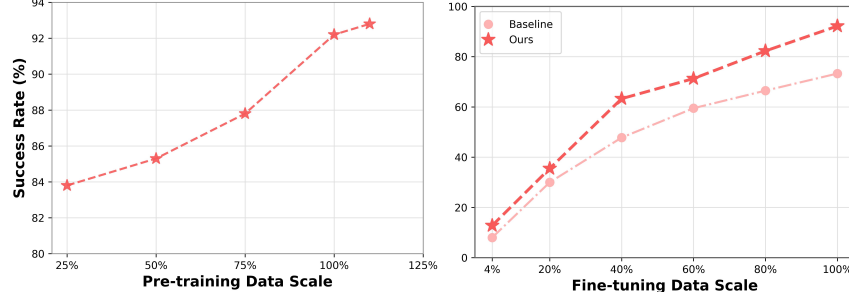


Figure 8: Influence of pretraining and fine-tuning data scale.

Table 7: Comparison of different parameter update strategies.

Method	SR(%)
Full Parameters	67.2 (↓ 25.0)
LN Fine-tuning	90.1 (↓ 2.1)
LoRA Fine-tuning	92.2

data. We also evaluate the sample efficiency of downstream policy learning by varying the number of expert demonstrations. LSA matches the peak performance of baseline methods using only 60% of the downstream expert demonstrations. This suggests that the multi-tiered 3D knowledge learned during pretraining reduces the burden of learning spatial and interaction cues from limited robot demonstrations.

C.2 Parameter Update Strategies

We compare different parameter update strategies during pretraining. As shown in Tab. 7, updating all parameters leads to a large performance drop. This may be caused by overfitting to the relatively limited pretraining data and forgetting the general visual priors of the initialized ViT. Fine-tuning only LayerNorm parameters performs much better, but still underperforms LoRA. Therefore, we adopt LoRA as the default parameter-efficient adaptation strategy, as it adapts the encoder to our 3D supervision while preserving the general visual representation ability of the pretrained backbone.

C.3 Pretraining Data Diversity

We further evaluate whether LSA benefits from more diverse human-video sources. As shown in Tab. 8, all settings use the same number of sampled clips, while the number of source videos varies. Pretraining on H2O [59] with around 100 source videos achieves 90.2%, while DexYCB with around 400 source videos improves the success rate to 92.2%. Using Ego4D [61] with around 1000 source videos further improves the result to 93.1%. These results suggest that LSA benefits from broader source-video diversity, indicating its potential to scale with more diverse human-video pretraining data.

C.4 Hyperparameter Selection

As shown in Fig. 9, we investigate the optimal configuration of three critical hyperparameters: the depth of attentive guidance R , the number of motion planning steps L , and the number of depth reconstruction decoder layers C . For attentive guidance, applying it to the first $R = 6$ layers achieves the highest success rate of 92.2%. Restricting guidance to only the first two layers ($R = 2$) lowers the success rate to 85.8%, because this limited supervision makes it difficult to identify a reliable target attention map and leaves other optimizable shallow layers unguided. In contrast, extending it too deeply ($R = 8$) reduces performance to 87.8%, as it limits the ability of deeper layers to learn abstract task semantics. For motion planning, $L = 3$ yields the best performance. A shorter horizon ($L = 1$) severely degrades the success rate to 78.3%, because such a short horizon lacks sufficient temporal context to represent meaningful hand motion. In contrast, a longer horizon ($L = 7$) reduces performance to 85.0%, since predicting farther-future trajectories becomes overly difficult

Table 8: Effect of pretraining data diversity. All settings use 20K sampled clips, while the number of source videos varies.

Data Source	#Videos	SR(%)
H2O	~ 100	90.2
DexYCB	~ 400	92.2
Ego4D	~ 1000	93.1

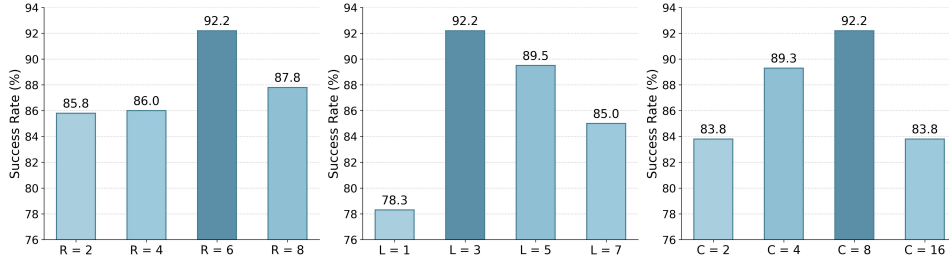


Figure 9: Impact of attentive guidance depth R , motion planning steps L , and the number of depth reconstruction decoder layers C .

and inaccurate. Finally, an 8-layer depth decoder ($C = 8$) best balances model capacity and training stability. A shallower decoder ($C = 2$) leads to a lower success rate of 83.8%, because its limited capacity is insufficient to reconstruct dense and accurate depth maps. In contrast, an overly deep decoder ($C = 16$) also reduces performance to 83.8%, as it introduces optimization difficulties that hinder overall training. Therefore, we adopt $R = 6$, $L = 3$, and $C = 8$ as our default setup.

C.5 Layer Selection

Tab. 9 presents an ablation on the layer configurations for representation alignment and depth reconstruction. While keeping the depth reconstruction fixed at the 6th layer, setting the alignment layer too shallow—such as the (1, 6) configuration—leads to a 2.7% drop in success rate, as the network has not yet extracted sufficient semantic features to effectively align with the high-dimensional implicit representations of the 3D FMs. Conversely, setting the alignment layer too deep, as seen in the (5, 6) setup, causes a 3.6% performance decrease by excessively compressing the subsequent feature space required for explicit structural perception.

Similarly, fixing the feature alignment at the 3rd layer and altering the depth reconstruction layer also significantly impacts performance. Introducing reconstruction too early, as in the (3, 4) configuration, results in a 3.9% decline in success rate. At this stage, the network has not fully assimilated the implicit 3D representations from the Lift phase, preventing it from effectively comprehending complex spatial layouts. Furthermore, delaying the reconstruction task to an excessively deep layer like (3, 8) triggers a severe performance drop of 7.6%. This is because representations become highly abstract as they approach the Act phase, focusing primarily on global motion trajectory planning. Forcing these deep layers to retain fine-grained, pixel-level geometric information inherently interferes with the action planning task.

In addition, reversing the order to (6, 3) results in a 2.9% drop. The model performs better when it first acquires a high-level 3D understanding (implicit) before attempting detailed geometric reconstruction (explicit). This suggests that the aligned 3D representations provide a necessary foundation for learning accurate 3D spatial layouts. Finally, forcing the model to perform both feature alignment and depth reconstruction at the exact same layer—such as in the (3, 3) or (6, 6) configurations—results in substantial performance drops of 6.4% and 9.9%, respectively. Since feature alignment and depth reconstruction are tasks of completely different granularities, forcing them to share the same feature space triggers severe feature interference.

Table 9: Ablation study on the selection of alignment and reconstruction layers.

Layer (Align, Recon)	SR(%)
(3, 6) (Ours)	92.2
<i>Varying Alignment Layer</i>	
(1, 6)	89.5 (↓ 2.7)
(5, 6)	88.6 (↓ 3.6)
<i>Varying Reconstruction Layer</i>	
(3, 4)	88.3 (↓ 3.9)
(3, 8)	84.6 (↓ 7.6)
<i>Reversed Order</i>	
(6, 3)	89.3 (↓ 2.9)
<i>Shared Layer</i>	
(3, 3)	85.8 (↓ 6.4)
(6, 6)	82.3 (↓ 9.9)

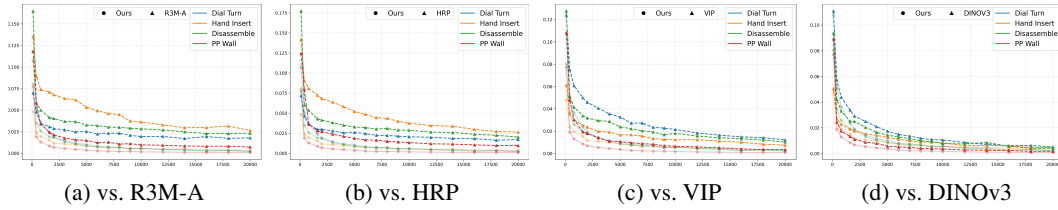


Figure 10: Comparison of training convergence between LSA and several baselines: (a) R3M-A, (b) HRP, (c) VIP, and (d) DINOv3.

C.6 Training Loss Curves

Fig. 10 presents the fine-tuning loss curves of different pretraining methods on MetaWorld. Our method exhibits the fastest and most stable convergence and reaches the lowest loss among all compared approaches. We attribute this advantage to the multi-tiered 3D supervision introduced during pretraining, which equips the policy with more generalizable spatial and motion-planning knowledge and thus enables faster adaptation to downstream manipulation tasks.

D Additional Visualizations

This section provides additional visualizations. We include representative simulation and real-world trajectories, as well as more attentive guidance visualizations on both human-object interaction and robotic manipulation scenes.

D.1 Simulation Trajectory Visualizations

Fig. 11 shows representative task trajectories in simulation. These examples qualitatively demonstrate that the policy pretrained by LSA can execute diverse manipulation behaviors across different simulated environments. The visualized trajectories cover both tabletop manipulation and dexterous manipulation settings, complementing the quantitative results reported in the main paper.

D.2 Real-World Trajectory Visualizations

Fig. 12 presents representative real-world trajectories executed by LSA. The examples correspond to the real-world tasks described in the main paper, including *Wipe PC*, *Pick Trash*, *Place Toy*, and *Pluck Flower*. These visualizations provide qualitative evidence that the learned representation can transfer from human-video pretraining to real-world robot manipulation.

749 D.3 More Attentive Guidance Visualizations

750 Fig. 13 presents additional visualization results of attentive guidance on human-object interaction
 751 scenes. The model learns to focus on hand-centric interaction regions, often covering both the hand
 752 and the interacting object. This indicates that the hand prior is not used to impose a fixed hand
 753 heatmap, but to mine an internally emerged interaction-aware attention target. Moreover, Fig. 14
 754 shows that such attention patterns can transfer to robotic manipulation scenes, where the model
 755 places stronger emphasis on task-relevant regions, such as the gripper and manipulated objects, while
 756 reducing attention to less relevant background regions.

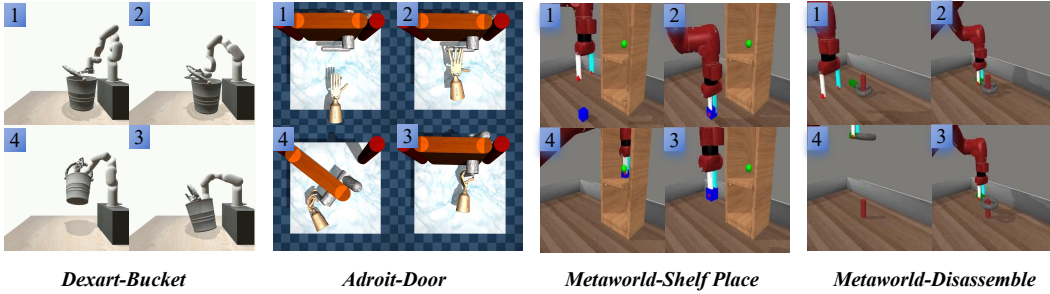


Figure 11: Visualization of representative simulation task trajectories. These examples illustrate the execution behavior of LSA in simulated manipulation environments.

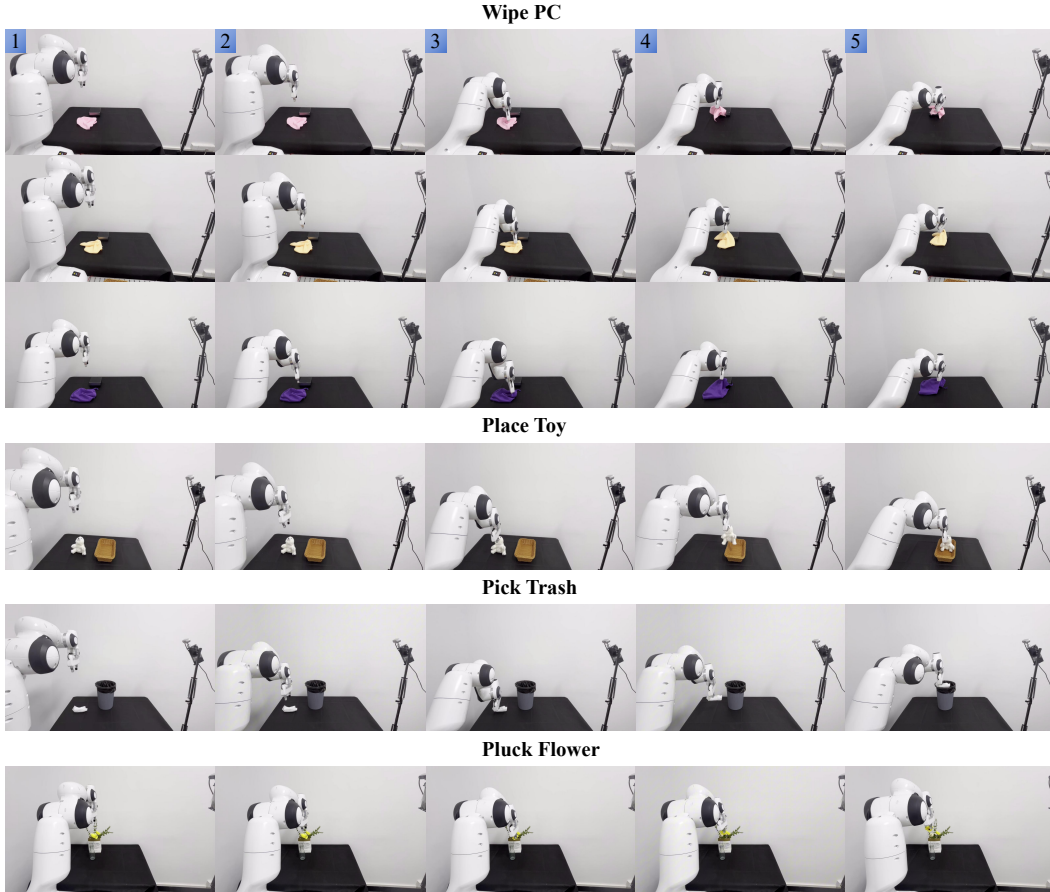


Figure 12: Visualization of representative real-world task trajectories. The examples show LSA executing real-world manipulation tasks with the Franka Research 3 robot.

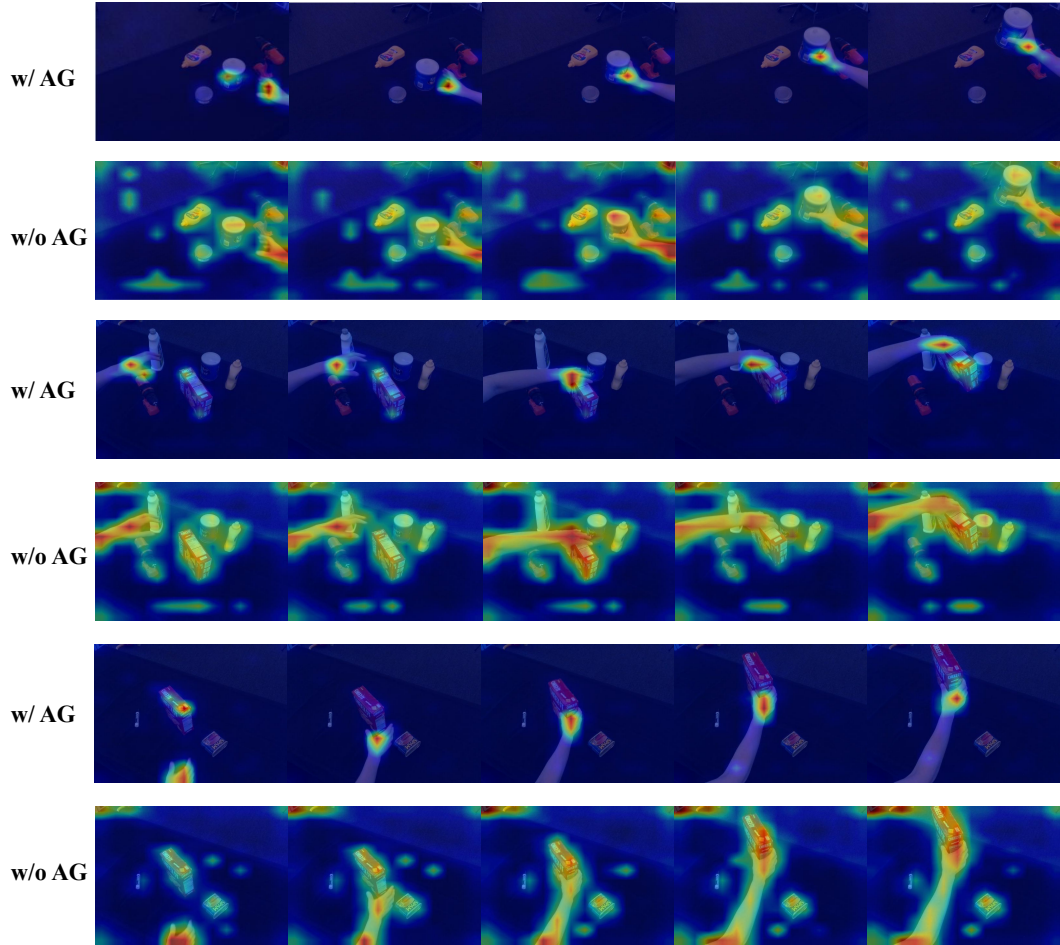


Figure 13: Additional visualization results of attentive guidance. The model learns to focus on hand-centric interaction regions, often covering both the hand and the interacting object.

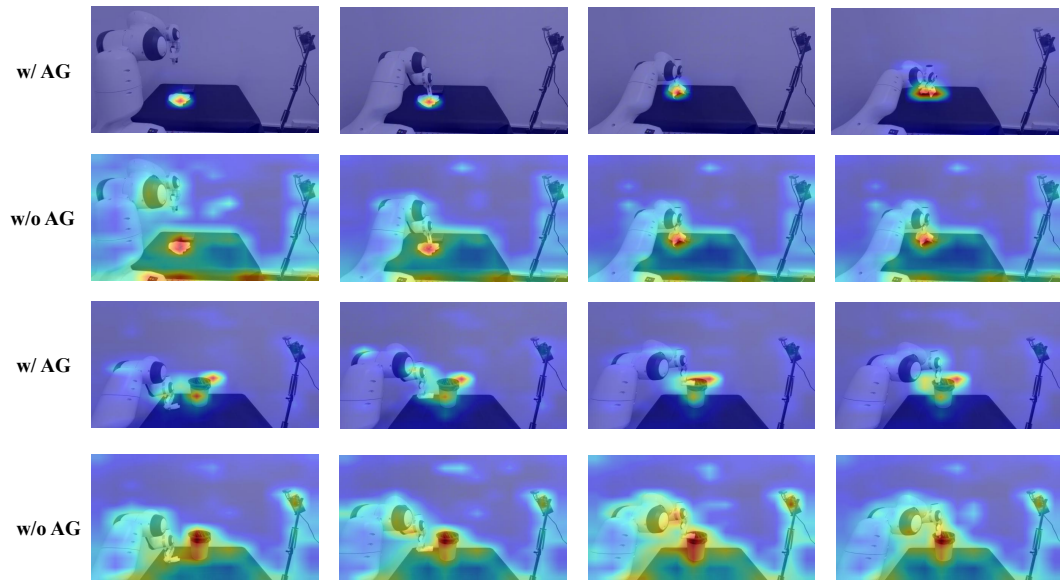


Figure 14: Additional visualization results of attentive guidance in robotic manipulation scenes. The model consistently places stronger attention on task-relevant regions, such as the gripper and manipulated objects, while reducing attention to background regions.

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